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The following code aggregates satellite raster data from WorldPop onto VTD's. Wrote this in python because the R version of the code takes forever to run. Adds raster pixels within VTD geometries from WorldPop rasters.

```
In [1]: import geopandas as gpd
        from rasterstats import zonal_stats
        import rasterio
        import pandas as pd
```

```
In [2]: sf24vtd = gpd.read_file("vtd24sf\\VTDs_24PG.shp")
```

```
In [5]: raster_crs = rasterio.open("satelliterasters/satellite2024cropped.tif").crs
```

```
In [6]: sf24vtd = sf24vtd.to_crs(raster_crs)
```

```
In [ ]: vap4_2024 = zonal_stats(
        sf24vtd,
        "satelliterasters/satellite2024cropped.tif",
        stats = ["sum"],
        geojson_out=False,
        nodata = 0
    )

vap4_2024 = [value for d in vap4_2024 for value in d.values()]
vap4_2024 = [0 if x is None else x for x in vap4_2024]
vap4_2024 = [0 if x < 0 else x for x in vap4_2024]
```

```
In [11]: vap4_2022 = zonal_stats(
        sf24vtd,
        "satelliterasters/satellite2022cropped.tif",
        stats = ["sum"],
        geojson_out=False,
        nodata = 0
    )

vap4_2022 = [value for d in vap4_2022 for value in d.values()]
vap4_2022 = [0 if x is None else x for x in vap4_2022]
vap4_2022 = [0 if x < 0 else x for x in vap4_2022]
```

```
In [12]: vap4_2020 = zonal_stats(
        sf24vtd,
        "satelliterasters/satellite2020cropped.tif",
        stats = ["sum"],
        geojson_out=False,
        nodata = 0
    )
```

```
vap4_2020 = [value for d in vap4_2020 for value in d.values()]
vap4_2020 = [0 if x is None else x for x in vap4_2020]
vap4_2020 = [0 if x < 0 else x for x in vap4_2020]
```

```
In [13]: vap4_2018 = zonal_stats(
          sf24vtd,
          "satelliterasters/satellite2018cropped.tif",
          stats = ["sum"],
          geojson_out=False,
          nodata = 0
        )

vap4_2018 = [value for d in vap4_2018 for value in d.values()]
vap4_2018 = [0 if x is None else x for x in vap4_2018]
vap4_2018 = [0 if x < 0 else x for x in vap4_2018]
```

```
In [15]: vap4 = pd.DataFrame({
          "vap4_2024": vap4_2024,
          "vap4_2022": vap4_2022,
          "vap4_2020": vap4_2020,
          "vap4_2018": vap4_2018
        })

vap4.to_csv("vap4.csv", index=False)
```

```
In [16]: vap4.head(n = 2)
```

```
Out[16]:
```

	vap4_2024	vap4_2022	vap4_2020	vap4_2018
0	3475.137939	3453.574707	3439.695312	3412.771973
1	3878.836426	3856.980225	3842.024170	3815.951660